



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 18-Apr-2023

Revision Number 3

## 1. Identification

### Product identifier

Product Name Silver Complex Solution

### Other means of identification

Catalog Number(s) 1610462, 1610462EDU

### Recommended use of the chemical and restrictions on use

Recommended use Laboratory chemicals

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
1329 Meyerside Drive  
Mississauga, ON L5T 1C9  
Canada

#### Technical Service

1-800-361-1808  
support@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

|                                   |             |
|-----------------------------------|-------------|
| Skin corrosion/irritation         | Category 2  |
| Serious eye damage/eye irritation | Category 2A |

### Label elements

#### **Warning**

#### **Hazard statements**

Causes skin irritation  
Causes serious eye irritation

**Precautionary Statements - Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Wear protective gloves/protective clothing/eye protection/face protection

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
If eye irritation persists: Get medical advice/attention

**Skin**

IF ON SKIN: Wash with plenty of water and soap  
If skin irritation occurs: Get medical advice/attention  
Take off contaminated clothing and wash it before reuse

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

Very toxic to aquatic life with long lasting effects. Very toxic to aquatic life.

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

| Chemical name  | CAS No    | Weight-% | Hazardous Material<br>Information Review Act<br>registry number<br>(HMIRA registry #) | Date HMIRA filed and<br>date exemption granted<br>(if applicable) |
|----------------|-----------|----------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Silver nitrate | 7761-88-8 | 1 - 2.5  | -                                                                                     |                                                                   |

**4. First-aid measures****Description of first aid measures****General advice**

Show this safety data sheet to the doctor in attendance.

**Inhalation**

Remove to fresh air. Get medical attention immediately if symptoms occur.

**Eye contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Keep eye wide open while rinsing. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention if irritation develops and persists. Do not rub affected area.

|                                           |                                                                                                                                                               |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin contact</b>                       | Wash off immediately with soap and plenty of water for at least 15 minutes. Get medical attention if irritation develops and persists.                        |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water. Never give anything by mouth to an unconscious person. Do NOT induce vomiting. Call a physician. |
| <b>Self-protection of the first aider</b> | Avoid contact with skin, eyes or clothing. Wear personal protective clothing (see section 8).                                                                 |

**Most important symptoms and effects, both acute and delayed**

|                 |                                                               |
|-----------------|---------------------------------------------------------------|
| <b>Symptoms</b> | May cause redness and tearing of the eyes. Burning sensation. |
|-----------------|---------------------------------------------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                           |                        |
|---------------------------|------------------------|
| <b>Note to physicians</b> | Treat symptomatically. |
|---------------------------|------------------------|

**5. Fire-fighting measures**

|                                                       |                                                                                                                                    |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| <b>Unsuitable extinguishing media</b>                 | No information available.                                                                                                          |
| <b>Specific hazards arising from the chemical</b>     | None known.                                                                                                                        |
| <b>Explosion data</b>                                 |                                                                                                                                    |
| <b>Sensitivity to mechanical impact</b>               | None.                                                                                                                              |
| <b>Sensitivity to static discharge</b>                | None.                                                                                                                              |
| <b>Special protective equipment for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

**6. Accidental release measures****Personal precautions, protective equipment and emergency procedures**

|                             |                                                                                                                        |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions</b> | Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. |
| <b>Other information</b>    | Refer to protective measures listed in Sections 7 and 8.                                                               |

**Methods and material for containment and cleaning up**

|                                |                                                       |
|--------------------------------|-------------------------------------------------------|
| <b>Methods for containment</b> | Prevent further leakage or spillage if safe to do so. |
| <b>Methods for cleaning up</b> | Pick up and transfer to properly labeled containers.  |

**7. Handling and storage****Precautions for safe handling**

|                                |                                                                                                                                                                                                                             |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Advice on safe handling</b> | Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse. |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store according to product and label instructions.

**8. Exposure controls/personal protection****Control parameters****Exposure Limits**

| Chemical name               | Alberta                     | British Columbia                                            | Ontario                     | Quebec                      |
|-----------------------------|-----------------------------|-------------------------------------------------------------|-----------------------------|-----------------------------|
| Silver nitrate<br>7761-88-8 | TWA: 0.01 mg/m <sup>3</sup> | TWA: 0.01 mg/m <sup>3</sup><br>STEL: 0.03 mg/m <sup>3</sup> | TWA: 0.01 mg/m <sup>3</sup> | TWA: 0.01 mg/m <sup>3</sup> |

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** If splashes are likely to occur, wear safety glasses with side-shields.

**Hand protection** Wear suitable gloves. Impervious gloves.

**Skin and body protection** Wear suitable protective clothing. Long sleeved clothing.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product. Avoid contact with skin, eyes or clothing.

**9. Physical and chemical properties****Information on basic physical and chemical properties**

**Physical state** Liquid  
**Appearance** aqueous solution  
**Color** colorless  
**Odor** Odorless  
**Odor threshold** No information available

| <b>Property</b>                               | <b>Values</b>     | <b>Remarks • Method</b> |
|-----------------------------------------------|-------------------|-------------------------|
| <b>pH</b>                                     |                   |                         |
| <b>Melting point / freezing point</b>         | No data available | None known              |
| <b>Boiling point / boiling range</b>          | > 100 °C / 212 °F |                         |
| <b>Flash point</b>                            | No data available | None known              |
| <b>Evaporation rate</b>                       | No data available | None known              |
| <b>Flammability (solid, gas)</b>              | No data available | None known              |
| <b>Flammability Limit in Air</b>              |                   | None known              |
| <b>Upper flammability or explosive limits</b> | No data available |                         |
| <b>Lower flammability or explosive limits</b> | No data available |                         |

|                              |                   |            |
|------------------------------|-------------------|------------|
| Vapor pressure               | No data available | None known |
| Vapor density                | No data available | None known |
| Relative density             | No data available | None known |
| Water solubility             | Miscible in water |            |
| Solubility in other solvents | No data available | None known |
| Partition coefficient        | No data available | None known |
| Autoignition temperature     | No data available | None known |
| Decomposition temperature    |                   | None known |
| Kinematic viscosity          | No data available | None known |
| Dynamic viscosity            | No data available | None known |

**Other information**

|                      |                 |
|----------------------|-----------------|
| Explosive properties | Not applicable. |
| Oxidizing properties | Not applicable. |
| Softening point      | Not applicable  |
| Molecular weight     | Not applicable  |
| VOC content          | Not applicable  |

**10. Stability and reactivity**

|                                    |                                                      |
|------------------------------------|------------------------------------------------------|
| Reactivity                         | No information available.                            |
| Chemical stability                 | Stable under normal conditions.                      |
| Possibility of hazardous reactions | None under normal processing.                        |
| Conditions to avoid                | None known based on information supplied.            |
| Incompatible materials             | Strong acids. Strong bases. Strong oxidizing agents. |
| Hazardous decomposition products   | None known based on information supplied.            |

**11. Toxicological information****Information on likely routes of exposure****Product Information**

|              |                                                                                                                                                   |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. May cause irritation of respiratory tract.                                      |
| Eye contact  | Specific test data for the substance or mixture is not available. Irritating to eyes. (based on components). Causes serious eye irritation.       |
| Skin contact | Specific test data for the substance or mixture is not available. Causes skin irritation. (based on components).                                  |
| Ingestion    | Specific test data for the substance or mixture is not available. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea. |

**Symptoms related to the physical, chemical and toxicological characteristics**

|          |                                                     |
|----------|-----------------------------------------------------|
| Symptoms | Redness. May cause redness and tearing of the eyes. |
|----------|-----------------------------------------------------|

**Acute toxicity****Numerical measures of toxicity**

|                                                                              |                 |
|------------------------------------------------------------------------------|-----------------|
| The following values are calculated based on chapter 3.1 of the GHS document |                 |
| ATEmix (oral)                                                                | 38,356.10 mg/kg |

ATEmix (dermal) 71,428.60 mg/kg

**Component Information**

| Chemical name               | Oral LD50            | Dermal LD50          | Inhalation LC50                     |
|-----------------------------|----------------------|----------------------|-------------------------------------|
| Silver nitrate<br>7761-88-8 | = 1173 mg/kg ( Rat ) | > 2000 mg/kg ( Rat ) | > 750 µg/m <sup>3</sup> ( Rat ) 4 h |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                                                                        |
|------------------------------------------|----------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Classification based on data available for ingredients. Irritating to skin.            |
| <b>Serious eye damage/eye irritation</b> | Classification based on data available for ingredients. Causes serious eye irritation. |
| <b>Respiratory or skin sensitization</b> | Based on available data, the classification criteria are not met.                      |
| <b>Germ cell mutagenicity</b>            | Based on available data, the classification criteria are not met.                      |
| <b>Carcinogenicity</b>                   | Based on available data, the classification criteria are not met.                      |

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Chemical name               | ACGIH | IARC     | NTP | OSHA |
|-----------------------------|-------|----------|-----|------|
| Silver nitrate<br>7761-88-8 | -     | Group 2A | -   | X    |

**Legend**

**IARC (International Agency for Research on Cancer)**

Group 2A - Probably Carcinogenic to Humans

**OSHA (Occupational Safety and Health Administration of the US Department of Labor)**

X - Present

|                                 |                                                                   |
|---------------------------------|-------------------------------------------------------------------|
| <b>Reproductive toxicity</b>    | Based on available data, the classification criteria are not met. |
| <b>STOT - single exposure</b>   | Based on available data, the classification criteria are not met. |
| <b>STOT - repeated exposure</b> | Based on available data, the classification criteria are not met. |
| <b>Target organ effects</b>     | Eyes, Skin, Nasal Septum.                                         |
| <b>Aspiration hazard</b>        | Based on available data, the classification criteria are not met. |

**12. Ecological information**

**Ecotoxicity** Very toxic to aquatic life with long lasting effects.

| Chemical name               | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                                | Toxicity to microorganisms | Crustacea                                                                                                                                 |
|-----------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Silver nitrate<br>7761-88-8 | -                    | LC50: 0.00512 - 0.00787mg/L (96h, Poecilia reticulata)<br>LC50: 0.009 - 0.02mg/L (96h, Lepomis macrochirus)<br>LC50: 0.0242 - 0.0484mg/L (96h, Lepomis macrochirus)<br>LC50: 0.05 - 0.07mg/L (96h, Lepomis macrochirus)<br>LC50: 0.001339 - 0.001637mg/L (96h, Oncorhynchus mykiss) | -                          | EC50: =0.0006mg/L (48h, Daphnia magna)<br>EC50: 0.0008 - 0.001mg/L (48h, Daphnia magna)<br>EC50: 0.0008 - 0.0011mg/L (48h, Daphnia magna) |

|  |  |                                                                                                                                                                                                                                                                                                                                                                              |  |  |
|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
|  |  | LC50: =0.0075mg/L (96h, Oncorhynchus mykiss)<br>LC50: 0.00839 - 0.1802mg/L (96h, Oncorhynchus mykiss)<br>LC50: 0.00452 - 0.00638mg/L (96h, Pimephales promelas)<br>LC50: 0.00181 - 0.00214mg/L (96h, Pimephales promelas)<br>LC50: 0.0064 - 0.0106mg/L (96h, Pimephales promelas)<br>LC50: =0.009mg/L (96h, Pimephales promelas)<br>LC50: =0.0027mg/L (96h, Cyprinus carpio) |  |  |
|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

**Other adverse effects** No information available.

### 13. Disposal considerations

#### Waste treatment methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

### 14. Transport information

**TDG** Not regulated

**DOT** Not regulated

### 15. Regulatory information

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**16. Other information**

|                                   |                                  |                       |                           |                                           |
|-----------------------------------|----------------------------------|-----------------------|---------------------------|-------------------------------------------|
| <b>NFPA</b>                       | <b>Health hazards</b> 2          | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Physical and chemical properties</b> - |
| <b>HMIS</b>                       | <b>Health hazards</b> 2          | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |
| <i>Chronic Hazard Star Legend</i> | <i>* = Chronic Health Hazard</i> |                       |                           |                                           |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 Japan GHS Classification  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 RTECS (Registry of Toxic Effects of Chemical Substances)  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 18-Apr-2023

**Revision Note** Significant changes throughout SDS. Review all sections.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**